

Memo 11: Case B Node/Core Speed-Test Matrix and Timing Analysis

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo defines a compact speed-test matrix for case B on Kestrel and summarizes the observed timing analysis. The goal is to estimate the present one-class FVCOM-MP runtime sensitivity to MPI rank count and node placement. The observed SEC/IT values, speedup calculations, and plots have been moved to `Analysis/analyze_memo_11_speed.py` so the analysis can be rerun without hand-editing the memo.

The current case-B run script uses:

- case name: `waterPACT_b`;
- run directory: `FVCOM_MP_test_run/RUN_b`;
- plastic input: `generic_plastic_b.inp`;
- sediment input: `mp_sediment_a_no_adv.inp`;
- current production-style Slurm layout: `nodes=4, ntasks-per-node=104`, total MPI ranks = 416.

Kestrel nodes are assumed to allow up to 104 MPI ranks per node for this test. The table below starts from one core, expands through useful within-node and across-node configurations, and then adds a C-series extension to search for the maximum practical long-run speedup.

Primary Speed-Test Matrix

Table 1: Recommended case-B speed-test combinations.

Run ID	Scaling role	Nodes	Tasks/node	Total ranks	Purpose
B01	single-core baseline	1	1	1	Absolute serial-like MPI baseline.
B02	small within-node	1	2	2	First parallel efficiency point.
B03	small within-node	1	4	4	Early scaling slope.
B04	small within-node	1	8	8	Useful low-cost debug point.
B05	medium within-node	1	16	16	Checks communication overhead onset.
B06	medium within-node	1	32	32	Mid-node scaling.
B07	half node	1	52	52	Half of one 104-core node.

Run ID	Scaling role	Nodes	Tasks/node	Total ranks	Purpose
B08	dense within-node	1	64	64	Common power-of-two-ish reference below full node.
B09	dense within-node	1	78	78	Three-quarter node occupancy.
B10	full node	1	104	104	Maximum single-node occupancy.
B11	spread-vs-pack check	2	52	104	Same total ranks as B10, but spread across two nodes.
B12	two full nodes	2	104	208	First full multi-node test.
B13	spread-vs-pack check	4	52	208	Same total ranks as B12, but spread across four nodes.
B14	three full nodes	3	104	312	Intermediate full-node scaling.
B15	current full setup	4	104	416	Matches the present case-B production-style layout.
B16	eight-node spread check	8	52	416	Same total ranks as B15, but spread across eight nodes.
B17	dense eight-node test	8	78	624	Intermediate eight-node occupancy.
B18	full eight-node test	8	104	832	Maximum eight-node occupancy.
B19	full five-node test	5	104	520	Checks whether full-node scaling continues past B15.
B20	full six-node test	6	104	624	Same total ranks as B17, but packed into six full nodes.
B21	full seven-node test	7	104	728	Near-eight-node full scaling point.
B22	six-node spread check	6	70	420	Similar rank count to B15/B16, spread over six nodes.
C01	ten-node spread check	10	52	520	Same total ranks as B19, but spread over ten nodes.
C02	ten-node moderate occupancy	10	64	640	Near the B17/B20 rank count with lower per-node occupancy.
C03	ten-node dense occupancy	10	78	780	Extends the best observed 78 ranks/node occupancy beyond eight nodes.
C04	twelve-node spread check	12	52	624	Same total ranks as B17/B20, but spread over twelve nodes.
C05	twelve-node moderate occupancy	12	64	768	Moderate twelve-node scaling point.
C06	twelve-node dense occupancy	12	78	936	Tests whether the B17 occupancy remains favorable past eight nodes.
C07	sixteen-node spread check	16	52	832	Same total ranks as B18, but spread over sixteen nodes.
C08	sixteen-node moderate occupancy	16	64	1024	Clean 1024-rank test for higher maximum speedup.
C09	sixteen-node dense occupancy	16	78	1248	Main candidate for a new maximum-speed configuration.
C10	sixteen-node full occupancy	16	104	1664	Diagnostic full-node test; checks whether full occupancy remains slower.

Priority Subset for Debug Queue

If the debug queue time or node limit is tight, the minimum useful subset is:

Run ID	Nodes	Tasks/node	Total ranks
B01	1	1	1
B04	1	8	8
B06	1	32	32
B10	1	104	104
B12	2	104	208
B15	4	104	416
B16	8	52	416
B20	6	104	624
B22	6	70	420
B18	8	104	832

This subset gives one single-core anchor, three within-node scaling points, and four multi-node points. If only one 8-node job is practical, keep B16 because it directly compares against B15 at the same total rank count.

Suggested Slurm Fields to Change

For each speed test, change only the Slurm resources and job name:

```
#SBATCH --partition=debug
#SBATCH --nodes=<Nodes>
#SBATCH --ntasks-per-node=<Tasks/node>
#SBATCH --job-name=WFPb_<Run ID>
```

Keep the executable command unchanged:

```
srun ./fvcom_TLMF --casename=waterPACT_b
```

For clean comparisons, use the same executable, same input files, same output settings, and same initial condition for every run. The speed-test output can be stored in separate Slurm log files; full NetCDF output is not needed if the debug run is only long enough to capture representative per-step timing.

Automated Timing Analysis

The raw observed timing table is no longer maintained inside this memo. To regenerate the SEC/IT plot, speedup plot, key comparison tables, and written interpretation, run:

```
python Analysis/analyze_memo_11_speed.py
```

from the repository root. The script writes the generated figures and the LaTeX analysis block included below.

Observed Timing Results

The observed SEC/IT values are stored in `Analysis/analyze_memo_11_speed.py`. That script computes the speedup relative to B01, parallel efficiency, same-rank spread/pack comparisons, and the figures below.

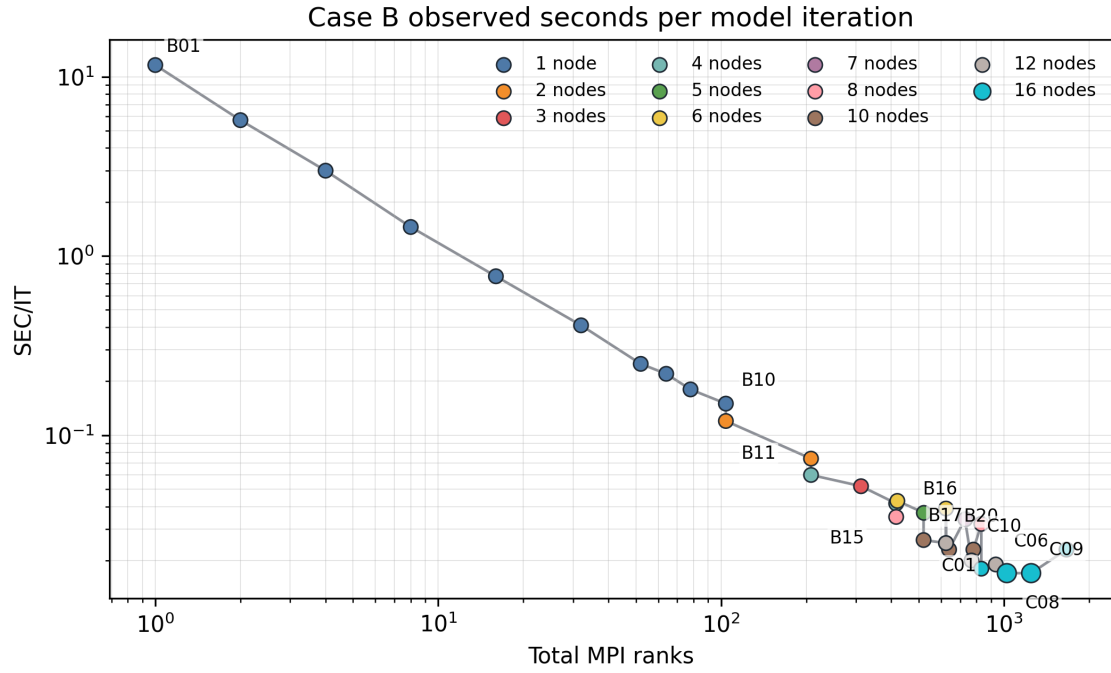


Figure 1: Observed case-B SEC/IT versus total MPI ranks. Lower values are faster; C08 and C09 are the fastest observed points.

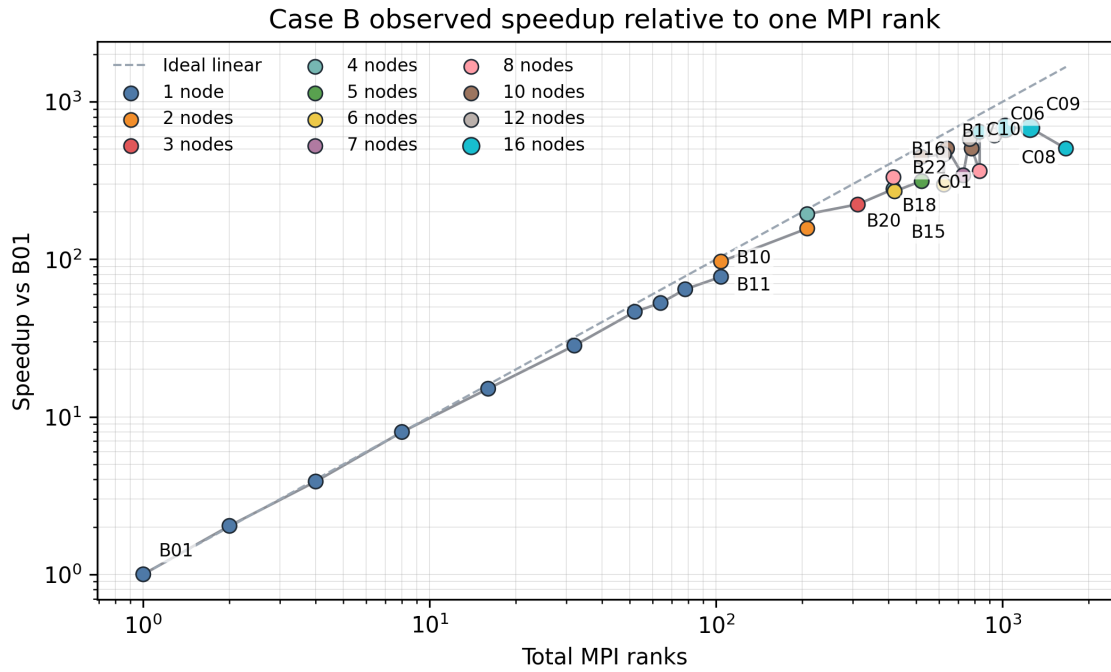


Figure 2: Observed speedup relative to the B01 one-rank baseline. The dashed line marks ideal linear speedup.

Table 2: Key observed timing configurations generated from the analysis script.

Run ID	Layout	Ranks	SEC/IT	Speedup	Efficiency
B15	4 x 104	416	0.0416	279.1	67.1%
B16	8 x 52	416	0.035	331.7	79.7%
B17	8 x 78	624	0.025	464.4	74.4%
B18	8 x 104	832	0.032	362.8	43.6%
C01	10 x 52	520	0.026	446.5	85.9%
C02	10 x 64	640	0.023	504.8	78.9%
C03	10 x 78	780	0.023	504.8	64.7%
C04	12 x 52	624	0.025	464.4	74.4%
C05	12 x 64	768	0.02	580.5	75.6%
C06	12 x 78	936	0.019	611.1	65.3%
C07	16 x 52	832	0.018	645.0	77.5%
C08	16 x 64	1024	0.017	682.9	66.7%
C09	16 x 78	1248	0.017	682.9	54.7%
C10	16 x 104	1664	0.023	504.8	30.3%

Table 3: Same-rank or near-diagnostic node-spread comparisons.

Rank case	Packed/reference	Spread/candidate	SEC/IT change
104 ranks	B10, 1 x 104	B11, 2 x 52	20.0% faster
208 ranks	B12, 2 x 104	B13, 4 x 52	18.9% faster
416 ranks	B15, 4 x 104	B16, 8 x 52	15.9% faster
520 ranks	B19, 5 x 104	C01, 10 x 52	29.7% faster
624 ranks	B20, 6 x 104	B17, 8 x 78	35.9% faster
624 ranks	B20, 6 x 104	C04, 12 x 52	35.9% faster
832 ranks	B18, 8 x 104	C07, 16 x 52	43.8% faster

Analysis Interpretation

The fastest observed SEC/IT is shared by **C08** and **C09**, both at 0.017 SEC/IT. Relative to the one-rank B01 baseline this is a 682.9x speedup, and relative to the current four-node B15 layout it is 59.1% lower in SEC/IT. Because C08 reaches the same timing with 1024 ranks instead of 1248, it has higher parallel efficiency and is the preferred default for long production-style runs at the present tracer count.

The C-series extension shows the benefit of moving beyond the previous eight-node tests. The ten-node C02/C03 timings reach 0.023 SEC/IT, the twelve-node C05/C06 timings improve to 0.020–0.019 SEC/IT, and the sixteen-node C08/C09 timings reach the current minimum of 0.017 SEC/IT.

For the near-416-rank operating point, **B16** is the best observed choice: 0.035 SEC/IT compared with 0.0416 SEC/IT for B15. The matched-rank checks show that spreading ranks across more nodes improves timing for 104, 208, 416, 520, 624, and 832 total ranks in these observations.

The eight-node full-occupancy run B18 is slower than B17 by 28.0%, even though B18 uses more ranks. This suggests that full node occupancy is not the preferred layout. The same pattern becomes clearer at sixteen nodes: C10, the 16 x 104 full-occupancy run, is 35.3% slower than C08. C09 ties C08 in SEC/IT, but its parallel efficiency is 17.9% lower because it uses 224 more ranks without reducing wall-clock time.

Node/Core Recommendation for Later Runs

For the current case-B timing set, use **C08** as the default long-run configuration when the queue allows sixteen nodes: 16 nodes, 64 ranks per node, 1024 total MPI ranks, and 0.017 SEC/IT. Use **C07** as the conservative sixteen-node option if the barotropic step or halo exchanges become noisy; it is only slightly slower at 0.018 SEC/IT and leaves more per-node headroom. Use **C09** only as a repeat-test candidate for the final multi-floc setup, because it currently gives no speed advantage over C08. Avoid **C10** for production unless a later physics configuration changes the scaling behavior.

When multiple floc tracers are enabled, the optimal point can shift because the tracer work increases, while the barotropic/external-mode work remains communication sensitive. If the added floc tracers make the run more compute-bound, C09 may become useful; if the barotropic step becomes the heavy or noisy component, lower per-node occupancy should be favored. The recommended final confirmation matrix is therefore C07, C08, C09, plus C06 as a lower-node fallback. Treat C08 as the present best balance of elapsed time, efficiency, and margin for communication-heavy barotropic work.